BLACK HOLES CARRY POSITIVE DISCRETE INFORMATION

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ABSTRACT. In type II string theory on $K3 \times T^2$, Sen's argument predicts that the single-centred black hole microstates carrying any fixed eigenvalue of a symplectic automorphism of $K3$ have positive index. We prove this for every symplectic automorphism and every electric charge when the magnetic charge satisfies $P^2 = 2$.

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1. INTRODUCTION

A supersymmetric black hole is a thermodynamic object, and its entropy counts microstates. In string theory one can count them exactly. For quarter-BPS dyons of type II string theory on $K3 \times T^2$, equivalently of the heterotic string on T^6 (on the heterotic side see also [29]), the protected index is a Fourier coefficient of the inverse of the Igusa cusp form Φ_{10} [15]. Sen observed that the index tells us more than a number: the near-horizon geometry of a single-centred black hole is spherically symmetric, the only states outside the horizon in a suitable duality frame are the zero modes of the broken supersymmetries, and the horizon states must therefore all be bosonic [33]. Once the zero modes are stripped off, the index of a single-centred black hole equals its degeneracy and is positive.[1]

Black holes carry more information than their degeneracy. When the compactification has a discrete symmetry g of order N that commutes with the supercharges, and the charges are g-invariant, the microstates split according to the eigenvalue $e^{2\pi i a/N}$ of g. Sen called the resulting refined counts the *discrete information* of the black hole and showed that the macroscopic side reproduces the twined indices $\mathrm{Tr}\, g^b(-1)^{2h}(2h)^6/6!$ through orbifolds of the near-horizon geometry [32]. The number of horizon states with eigenvalue $e^{2\pi i a/N}$ is

$$S_a = \frac{1}{N} \sum_{b=0}^{N-1} e^{-2\pi i a b/N}\, d_{g^b}, \tag{1}$$

where d_{g^b} is the single-centred index twined by g^b and $d_{g^0} = d$ is the ordinary one. Sen's sign argument applies to each eigenspace separately, since the same spherically symmetric attractor geometry computes all the twined indices. Hence S_a counts bosonic horizon states, and the argument predicts that it is a positive integer [22, Section 1]. Govindarajan, Samanta, Shanmugapriya and Virmani formulated this prediction and tested it for small charges and for $N = 2, 3, 4$ [22]. As far as we know, positivity of the discrete information has not been proved for any infinite family of charges.

[1]The sign convention is $-B_6 = d_{\mathrm{hor}}$, where B_6 is the sixth helicity supertrace. The argument is in Sections 1 and 2 of [33]; the relation between index and degeneracy is derived from the quantum entropy function in [12], and the entropy function formalism is reviewed in [31]. In frames where the classical solution carries hair one must first remove it; in the frame used here no hair besides the fermion zero modes is present, see [9] and [22, Section 1].

For type II string theory on $K3 \times T^2$ the symmetries in question are the symplectic automorphisms of $K3$. Nikulin and Mukai showed that a finite symplectic automorphism has order $N \leq 8$ and falls into one of seven types, determined by its action on the cohomology of $K3$ [26, 25]. In the language of Mathieu moonshine these are the classes $2A$, $3A$, $4B$, $5A$, $6A$, $7A$ and $8A$ of M_{24} [16, 10, 19, 18]. The twined partition function $1/\Phi_g$ is the inverse of a Siegel modular form attached to g [13, 32, 21, 20, 10, 28, 27].

Write $n = Q^2/2$, $\ell = Q \cdot P$, $m = P^2/2$ and $D = 4mn - \ell^2$. Our main result is the following.

Theorem A. *Let g be a symplectic automorphism of finite order N of a $K3$ surface, and let (Q, P) be a g-invariant dyon with $P^2 = 2$ and $Q^2P^2 - (Q \cdot P)^2 > 0$. Then*

$$S_a(Q, P) > 0 \qquad \text{for every } a \in \mathbb{Z}/N\mathbb{Z}.$$

For $N = 1$ this is the theorem of Bringmann and Murthy [6]. The proof for $N \geq 2$ rests on two results that have independent interest. The first is an explicit formula for the generating function of the twined single-centred indices. Let $\phi_{-2,1}$ and $\phi_{0,1}$ be the generators of the ring of weak Jacobi forms of index one, let $\mathcal{H}(\tau, z) = \sum H(4n - \ell^2) q^n y^\ell$ be the generating function of the Hurwitz class numbers, and let η_g be the eta product of the cycle shape of g (Table 1).

Theorem B. *For every symplectic automorphism g there are modular forms $f_g \in M_4(\Gamma_0(N))$ and $h_g \in M_2(\Gamma_0(N))$, given explicitly in Table 3, such that the twined single-centred indices of magnetic charge $P^2 = 2$ are*

$$d_g(n, \ell) = (-1)^{\ell+1} c_g^F(n, \ell), \qquad \sum_{n,\ell} c_g^F(n, \ell)\, q^n y^\ell = \frac{f_g\, \phi_{-2,1} + h_g\, \phi_{0,1} - 2\, d_g(2)\, \mathcal{H}}{\eta_g},$$

where $d_g(2)$ is the coefficient of q^2 in q/η_g.

For $g = 1$ one has $\eta_1 = \Delta$, the discriminant function whose coefficients are Ramanujan's $\tau(n)$ [35, 2], and $f_1 = 3E_4$, $h_1 = 0$, $d_1(2) = 324$, so that Theorem B reduces to the formula $\Delta\, \psi_1^F = 3E_4\phi_{-2,1} - 648\, \mathcal{H}$ of Dabholkar, Murthy and Zagier used in [6]. The second result is a quantitative domination of the twined indices by the untwined one.

Theorem C. *For every symplectic automorphism g of order $N \geq 2$ and every $D > 0$,*

$$\sum_{b=1}^{N-1} \left| d_{g^b}(D) \right| < d(D).$$

Theorem A follows at once: by (1), $NS_a \geq d - \sum_{b \neq 0} |d_{g^b}| > 0$. Theorem C is much stronger than what is needed for large charges. The single-centred index grows like $e^{\pi\sqrt{D}}$, and numerically the ratio of the two sides of Theorem C decays roughly exponentially in $\sqrt{D}$ (Figure 2); its largest value is 0.052, attained for g of type $6A$ at $D = 4$, where the counts NS_a differ from d by up to 5.2 per cent (Figure 3). The work lies in proving the inequality for every D. A coefficientwise majorant loses most of the numerical room, and the coefficient inequality to which it reduces the problem holds with a margin of only one third at its tightest point.

The proof of Theorem B has three steps. The twining genus of $K3$ is computed from the fixed points of g by the holomorphic Lefschetz formula (Appendix A). The coefficient of p^1 in $1/\Phi_g$ is a meromorphic Jacobi form $\psi_{g,1}$ with a double pole at $z = 0$, and subtracting its polar part, an Appell–Lerch sum, leaves a mock Jacobi form whose coefficients are exactly the indices defined by Sen's attractor contour (Section 3). The remaining weak Jacobi form is identified by the structure theorem for weak Jacobi forms of index one and Sturm's bound (Section 4). The proof of Theorem C compares coefficients of explicit q-series. All twined terms are bounded coefficientwise by a single series divided by $(q; q)_\infty^{24}$, and a lemma on coloured partitions reduces

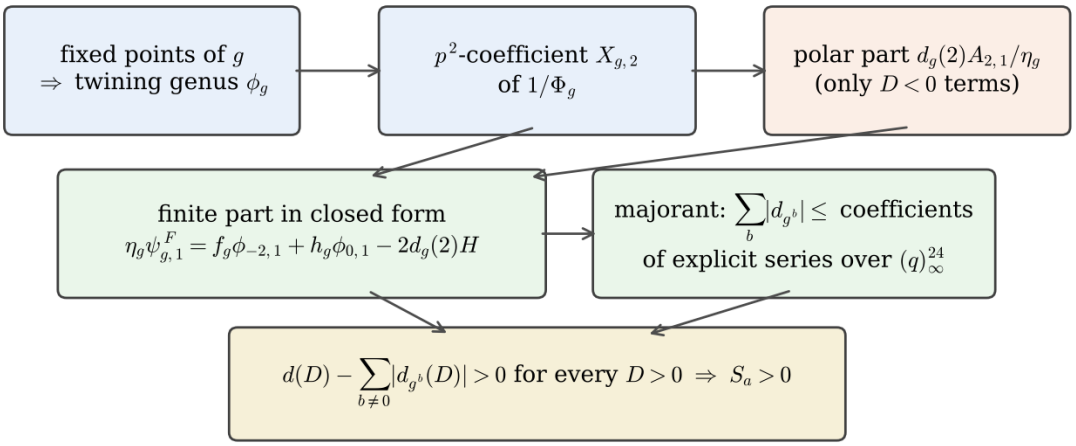


FIGURE 1. Structure of the proof. The polar part of the Fourier–Jacobi coefficient is an Appell–Lerch sum which does not contribute in the attractor region; the finite part has the closed form of Theorem B, and a coefficientwise majorant gives Theorem C.

the comparison with the untwined index to a coefficient inequality for one q-series per type and residue class. That inequality is proved analytically for $m \geq 11000$ and by exact rational arithmetic below (Sections 5 and 6). The structure of the argument is shown in Figure 1.

Notation. Throughout, $\tau \in \mathbb{H}$, $z \in \mathbb{C}$, $q = e^{2\pi i\tau}$, $y = e^{2\pi iz}$ and $(q;q)_\infty = \prod_{n\geq 1}(1-q^n)$; we write $(q)_\infty$ for short. For a power series $F(q) = \sum a_m q^m$ we write $[q^m]F = a_m$, and $|F| = \sum |a_m| q^m$. For two series with real coefficients, $F \preceq G$ means $[q^m]F \leq [q^m]G$ for all m. If $0 \preceq F \preceq G$ and $0 \preceq F' \preceq G'$ then $FF' \preceq GG'$. We write $\sigma_k(m) = \sum_{d|m} d^k$, $\sigma = \sigma_1$, and $p_k(m) = [q^m](q)_\infty^{-k}$ for the number of k-coloured partitions. The Eisenstein series are $E_2 = 1 - 24\sum \sigma(m)q^m$, $E_4 = 1 + 240\sum \sigma_3(m)q^m$, and for $d \geq 2$ we put

$$\phi^{(d)}(\tau) = \frac{d\,E_2(d\tau) - E_2(\tau)}{d-1} \in M_2(\Gamma_0(d)).$$

2. SYMPLECTIC AUTOMORPHISMS AND THE TWINED PARTITION FUNCTION

2.1. **The seven types.** Let X be a complex $K3$ surface and g an automorphism of finite order $N > 1$ preserving the holomorphic two-form. Nikulin proved that g has finitely many fixed points, that $N \leq 8$, and that the number of fixed points depends only on N: it is $8, 6, 4, 4, 2, 3, 2$ for $N = 2, \ldots, 8$ [26]. At a fixed point g acts on the tangent space by $\mathrm{diag}(\zeta, \zeta^{-1})$ with ζ a primitive root of unity of order dividing N, and the trace of g on $H^*(X, \mathbb{Q})$ equals the number of fixed points. Mukai showed that the action of g on $H^*(X, \mathbb{Q}) \cong \mathbb{Q}^{24}$ is that of an element of $M_{23} \subset M_{24}$ with at least five orbits on the 24 points [25]; with Nikulin's fixed-point count this determines the cycle shape, and hence the conjugacy class in M_{24}, from N. The seven types are listed in Table 1. Symplectic involutions and the Kummer and Enriques surfaces attached to them are discussed in [7], and $K3$-fibred Calabi–Yau threefolds, on which such automorphisms can act fibrewise, in [3].

TABLE 1. Symplectic automorphisms of $K3$. The cycle shape $\prod a^{m_a}$ is that of the permutation action on $\mathbb{Q}^{24}$; the eta product is $\eta_g(\tau) = \prod_a \eta(a\tau)^{m_a}$, and χ_g is the number of fixed points. The local rotation angles α (with $\zeta = e^{2\pi i\alpha}$) are listed with multiplicity. The last column is the class of g^2.

type	N	cycle shape	χ_g	rotation angles at the fixed points	g^2
$2A$	2	$1^8 2^8$	8	$\frac{1}{2}$ (8 times)	$1A$
$3A$	3	$1^6 3^6$	6	$\frac{1}{3}$ (6 times)	$3A$
$4B$	4	$1^4 2^2 4^4$	4	$\frac{1}{4}$ (4 times)	$2A$
$5A$	5	$1^4 5^4$	4	$\frac{1}{5}, \frac{1}{5}, \frac{2}{5}, \frac{2}{5}$	$5A$
$6A$	6	$1^2 2^2 3^2 6^2$	2	$\frac{1}{6}, \frac{1}{6}$	$3A$
$7A$	7	$1^3 7^3$	3	$\frac{1}{7}, \frac{2}{7}, \frac{3}{7}$	$7A$
$8A$	8	$1^2 2\, 4\, 8^2$	2	$\frac{1}{8}, \frac{3}{8}$	$4B$

In each case $\sum_a a\, m_a = 24$, so $\eta_g(\tau) = q\, P_g(q)$ with $P_g(q) = \prod_a (q^a;q^a)_\infty^{m_a}$. We put

$$d_g(k) = [q^k]\, \frac{1}{P_g(q)}.$$

For $k = 2$ these numbers are $324, 52, 27, 16, 14, 7, 9, 6$ for the classes $1A, 2A, \ldots, 8A$. The total number of cycles $\sum_a m_a$ is at most 16, and this will matter in Section 5. Since the type of g^b depends only on $N/\gcd(b,N)$, the powers of g are determined by the last column of Table 1 and by g^3 of type $2A$ for g of type $6A$.

2.2. **Twining genera.** The twining genus $\phi_g(\tau, z)$ of X is the trace of g on the Ramond sector of the $K3$ sigma model weighted as in the elliptic genus; equivalently it is the g-equivariant elliptic genus of X. For $g = 1$ it is $2\phi_{0,1}$. For every g in Table 1, ϕ_g is a weak Jacobi form of weight 0 and index 1 for $\Gamma_0(N)$ with trivial character [10, 19].

Proposition 2.1. *With χ_g and $T_g \in M_2(\Gamma_0(N))$ as in Table 2,*

$$\phi_g = \frac{\chi_g}{12}\, \phi_{0,1} + T_g\, \phi_{-2,1}.$$

The proof is in Appendix A. The formula agrees with the twining genera of Mathieu moonshine. With $\Lambda_d = \frac{d}{24}\big(dE_2(d\tau) - E_2(\tau)\big) = \frac{d(d-1)}{24}\phi^{(d)}$, the forms of Table 2 are $16\Lambda_2$, $6\Lambda_3$, $4(\Lambda_4 - \Lambda_2)$, $2\Lambda_5$, $2(\Lambda_6 - \Lambda_3 - \Lambda_2)$, Λ_7 and $\Lambda_8 - \Lambda_4$, which is the form in which the weight-two parts of the twining genera appear in [10, 19].[2]

TABLE 2. The weight-two forms T_g of Proposition 2.1.

g	$2A$	$3A$	$4B$	$5A$	$6A$	$7A$	$8A$
T_g	$\frac{4}{3}\phi^{(2)}$	$\frac{3}{2}\phi^{(3)}$	$-\frac{1}{3}\phi^{(2)} + 2\phi^{(4)}$	$\frac{5}{3}\phi^{(5)}$	$-\frac{1}{6}\phi^{(2)} - \frac{1}{2}\phi^{(3)} + \frac{5}{2}\phi^{(6)}$	$\frac{7}{4}\phi^{(7)}$	$-\frac{1}{2}\phi^{(4)} + \frac{7}{3}\phi^{(8)}$

[2]Our proof does not use moonshine: Proposition 2.1 is derived from the fixed points alone, and the comparison with [10, 19] is only a consistency check.

2.3. **The twined partition function.** Write $\phi_g = \sum c_g(n,\ell) q^n y^\ell$ and put $p = e^{2\pi i \sigma}$ for a third variable $\sigma \in \mathbb{H}$. The generating function of the twined indices of quarter-BPS dyons is $1/\Phi_g(\tau, z, \sigma)$, where [13, 32, 10, 27]

$$\frac{1}{\Phi_g} = \frac{1}{p\,\eta_g(\tau)\,\phi_{-2,1}(\tau,z)} \exp\Big(\sum_{k\ge1}\sum_{r\ge1}\sum_{n\ge0}\sum_{\ell\in\mathbb{Z}} \frac{1}{k}\, c_{g^k}(rn,\ell)\,\big(p^r q^n y^\ell\big)^k\Big). \tag{2}$$

This is the second-quantised twining genus of Dijkgraaf, Moore, Verlinde and Verlinde [14] multiplied by the contribution of the centre of mass and of the Kaluza–Klein monopole. Raum and Persson–Volpato proved that Φ_g is a Siegel modular form for a level-N subgroup of $\mathrm{Sp}_4(\mathbb{Z})$ [28, 27]; we do not use modularity in the variable σ.

Expanding (2) in powers of p gives $1/\Phi_g = \sum_{m\ge-1} \psi_{g,m}\, p^m$ with $\psi_{g,m} = X_{g,m+1}/(\eta_g \phi_{-2,1})$, where $X_{g,0} = 1$, $X_{g,1} = \phi_g$ and

$$X_{g,2}(\tau,z) = \tfrac12\,\phi_g(\tau,z)^2 + \big(U_2\phi_g\big)(\tau,z) + \tfrac12\,\phi_{g^2}(2\tau,2z), \qquad U_2\phi = \sum_{n,\ell} c(2n,\ell)\, q^n y^\ell. \tag{3}$$

The three terms come from $(k,r) = (1,1)$ squared, $(k,r) = (1,2)$, and $(k,r) = (2,1)$. The function $\psi_{g,m}$ is the generating function of twined indices of magnetic charge $P^2/2 = m$. In this paper $m = 1$, and we abbreviate $\psi_g = \psi_{g,1}$.

2.4. **The attractor contour.** The index of a dyon with charges (n,ℓ,m) is a Fourier coefficient of $1/\Phi_g$, but the coefficient depends on the region in which the meromorphic function is expanded, because of wall crossing. Sen's attractor contour selects the expansion adapted to the attractor values of the moduli, and the resulting coefficients are the indices of single-centred black holes [30, 11, 8]. For the Fourier–Jacobi coefficient $\psi_{g,m}$ this means: expand $\psi_{g,m}$ in powers of q and y in the strip of z containing the point $z = -\ell\tau/(2m)$ and take the coefficient of $q^n y^\ell$.[3] When $\ell/(2m)$ is an integer the point lies on a line of poles and either adjacent strip may be used; Lemma 3.1 shows that the answer is the same.

Definition 2.2. Let $D = 4n - \ell^2 > 0$. The *twined single-centred index* of a g-invariant dyon with $Q^2/2 = n$, $Q\cdot P = \ell$ and $P^2/2 = 1$ is

$$d_g(n,\ell) = (-1)^{\ell+1}\,\big[q^n y^\ell\big]\,\psi_g\big|_S,$$

where $\psi_g|_S$ denotes the Fourier expansion of ψ_g in a strip $S = \{-(s+1)\operatorname{Im}\tau < \operatorname{Im} z < -s\operatorname{Im}\tau\}$, $s\in\mathbb{Z}$, whose closure contains $-\ell\tau/2$. We write $d = d_1$.

The sign $(-1)^{\ell+1}$ removes the contribution of the fermion zero modes; with it $d_g = -B_6^g$ in the notation of [22]. The discrete information is $S_a = \frac1N \sum_b e^{-2\pi i ab/N} d_{g^b}(n,\ell)$.

3. The twined single-centred index

3.1. **The polar part.** Let

$$A_{2,1}(\tau,z) = \sum_{s\in\mathbb{Z}} \frac{q^{s^2+s} y^{2s+1}}{(1-q^s y)^2}.$$

[3] The attractor value of $\operatorname{Im} z/\operatorname{Im}\tau$ is $-\ell/(2m)$; see [30, Section 8] and [8, Section 2]. In [22] the same prescription appears as the condition $0 \le Q\cdot P \le \min(Q^2, P^2)$ together with an expansion in $e^{-2\pi i v}$; for $m = 1$ this is the strip $-\operatorname{Im}\tau < \operatorname{Im} z < 0$, and Lemma 3.1 below shows that it gives the same numbers.

This is the Appell–Lerch sum of [6, Section 3] for $m = 1$. It is meromorphic in z with double poles exactly at $z \in \mathbb{Z} + \tau\mathbb{Z}$, and near $z = 0$ the term $s = 0$ gives $A_{2,1} = y/(1-y)^2 + O(1) = 1/(2\pi i z)^2 + O(1)$.[4]

Lemma 3.1. *Let $s_0 \in \mathbb{Z}$ and expand $A_{2,1}$ in the strip $S_{s_0} = \{-(s_0+1)\operatorname{Im}\tau < \operatorname{Im} z < -s_0 \operatorname{Im}\tau\}$. Every monomial $q^n y^\ell$ that occurs has $4n - \ell^2 = -(k+1)^2$ for some integer $k \geq 0$, and no monomial with $\ell \in \{2s_0, 2s_0+1, 2s_0+2\}$ occurs.*

Proof. In S_{s_0} one has $|q^s y| < 1$ if and only if $s \operatorname{Im}\tau + \operatorname{Im} z > 0$, that is, if and only if $s \geq s_0 + 1$. For such s,

$$\frac{q^{s^2+s} y^{2s+1}}{(1-q^s y)^2} = \sum_{k \geq 0} (k+1)\, q^{s^2+s+ks}\, y^{2s+1+k},$$

and the exponents satisfy $4(s^2+s+ks) - (2s+1+k)^2 = -(k+1)^2$ and $\ell = 2s+1+k \geq 2s_0+3$. For $s \leq s_0$ one has $|q^s y| > 1$, and

$$\frac{q^{s^2+s} y^{2s+1}}{(1-q^s y)^2} = \sum_{k \geq 0} (k+1)\, q^{s^2+s-s(k+2)}\, y^{2s+1-(k+2)},$$

with $4(s^2 - s(k+1)) - (2s-k-1)^2 = -(k+1)^2$ and $\ell = 2s-k-1 \leq 2s_0 - 1$. Hence every exponent ℓ lies outside $\{2s_0, 2s_0+1, 2s_0+2\}$. □

If ℓ is odd, the attractor point $-\ell\tau/2$ lies in the strip $S_{(\ell-1)/2}$, and $\ell = 2s_0 + 1$. If ℓ is even, it lies on the boundary of $S_{\ell/2}$ and $S_{\ell/2-1}$, and ℓ equals $2s_0$ in the first and $2s_0 + 2$ in the second. In all cases ℓ is one of the three excluded exponents of the strip used in Definition 2.2.

3.2. **The finite part.** Let $\phi_{-2,1} = A$ and $\phi_{0,1} = B$ be the weak Jacobi forms

$$A = (y - 2 + y^{-1}) \prod_{n \geq 1} \frac{(1-q^n y)^2 (1-q^n y^{-1})^2}{(1-q^n)^4}, \qquad B = 4 \sum_{i=2}^{4} \frac{\vartheta_i(\tau, z)^2}{\vartheta_i(\tau, 0)^2},$$

so that $A = (2\pi i z)^2 + O(z^4)$ and $B(\tau, 0) = 12$ [17, Section 9]. Let

$$\mathcal{H}(\tau, z) = \sum_{n \geq 0} \sum_{\ell \in \mathbb{Z}} H(4n - \ell^2)\, q^n y^\ell,$$

where $H(0) = -\frac{1}{12}$, $H(N)$ for $N > 0$ is the Hurwitz class number (the number of $\mathrm{SL}_2(\mathbb{Z})$-classes of positive definite binary quadratic forms of discriminant $-N$, those equivalent to multiples of $x^2 + y^2$ or $x^2 + xy + y^2$ counted with weight $\frac{1}{2}$ or $\frac{1}{3}$), and $H(N) = 0$ if $N < 0$ or $N \equiv 1, 2 \pmod 4$ [24, 35].

Proposition 3.2 (Dabholkar–Murthy–Zagier)**.** *As meromorphic functions,*

$$\frac{B^2 - E_4 A^2}{144\, A} = A_{2,1} - 2\,\mathcal{H}.$$

[4] The other terms are holomorphic at $z = 0$ since $q^s \neq 1$ for $s \neq 0$. The leading behaviour of $y/(1-y)^2$ at $y = 1$ is $1/(2\pi i z)^2$, and $y/(1-y)^2$ is even in z, so there is no simple pole.

This identity is the case $m = 1$ of the decomposition of B^{m+1}/A into polar and finite parts; it is stated as $(B^2/A)^F = E_4 A - 288\,\mathcal{H}$ in [6, Section 3], where $(B^2/A)^F = B^2/A - 144\,A_{2,1}$.[5] We now derive the twined analogue.

Proposition 3.3. *Put* $W_g = \big(X_{g,2} - \frac{d_g(2)}{144}(B^2 - E_4 A^2)\big)/A$. *Then* W_g *is holomorphic on* $\mathbb{H}\times\mathbb{C}$, *and*

$$\eta_g\,\psi_g = W_g - 2\,d_g(2)\,\mathcal{H} + d_g(2)\,A_{2,1}. \tag{4}$$

Proof. The identity (4) is the definition of W_g combined with Proposition 3.2, since $\eta_g\psi_g = X_{g,2}/A$. For holomorphy, A vanishes exactly on $\mathbb{Z}+\tau\mathbb{Z}$, to order two [17, Theorem 9.3], and the numerator $N_g = X_{g,2} - \frac{d_g(2)}{144}(B^2 - E_4A^2)$ satisfies the elliptic transformation law of index two, $N_g(\tau, z+\lambda\tau+\mu) = q^{-2\lambda^2}y^{-4\lambda}N_g(\tau,z)$; so the order of vanishing of N_g is the same at all points of $\mathbb{Z}+\tau\mathbb{Z}$, and it suffices to show that N_g vanishes to order two at $z=0$. All functions involved are even in z, so N_g has no linear term. At $z=0$ we have $\phi_g(\tau,0) = \chi_g$ and $\phi_{g^2}(\tau,0) = \chi_{g^2}$ (the Euler characteristic of the fixed locus), and $(U_2\phi_g)(\tau,0) = \sum_n\big(\sum_\ell c_g(2n,\ell)\big)q^n = \chi_g$. Hence

$$X_{g,2}(\tau,0) = \tfrac{1}{2}\chi_g^2 + \chi_g + \tfrac{1}{2}\chi_{g^2}.$$

On the other hand, at $z=0$ only the terms with $n=0$ survive in (2), because $\sum_\ell c_{g^k}(j,\ell) = \chi_{g^k}\delta_{j,0}$, so $X_{g,2}(\tau,0)$ is the coefficient of p^2 in

$$\exp\Big(\sum_{r\ge1}\sum_{k\ge1}\frac{1}{k}\,\chi_{g^k}\,p^{rk}\Big) = \prod_{r\ge1}\prod_a (1-p^{ar})^{-m_a} = \frac{1}{P_g(p)}.$$

The middle identity follows from $\chi_{g^k} = \sum_{a|k} a\,m_a$, the number of points of $\{1,\dots,24\}$ fixed by g^k, which gives $\sum_k \frac{1}{k}\chi_{g^k}x^k = -\sum_a m_a\log(1-x^a)$. Therefore $X_{g,2}(\tau,0) = [p^2]\,P_g(p)^{-1} = d_g(2)$, which equals $\frac{d_g(2)}{144}B(\tau,0)^2$. So $N_g(\tau,0) = 0$. □

Theorem 3.4. *Let* $\psi_g^F = (W_g - 2\,d_g(2)\,\mathcal{H})/\eta_g = \sum c_g^F(n,\ell)q^n y^\ell$. *Then* $c_g^F(n,\ell) = C_g^F(4n-\ell^2)$ *depends only on the discriminant, and for every* (n,ℓ) *with* $D = 4n-\ell^2 > 0$,

$$d_g(n,\ell) = (-1)^{\ell+1}\,C_g^F(D) = (-1)^{D+1}\,C_g^F(D).$$

Proof. By Proposition 3.3, $\psi_g = \psi_g^F + d_g(2)A_{2,1}/\eta_g$. The function ψ_g^F is holomorphic in z, so its Fourier expansion is the same in every strip. The second term is $d_g(2)\,q^{-1}P_g(q)^{-1}A_{2,1}$, and multiplication by a power series in q does not change the exponent of y. By Lemma 3.1 the expansion of $A_{2,1}$ in the strip of Definition 2.2 contains no monomial with exponent ℓ of y, so the coefficient of $q^n y^\ell$ of $\psi_g|_S$ equals $c_g^F(n,\ell)$. That c_g^F depends only on $4n-\ell^2$ follows from the elliptic invariance of W_g (Proposition 3.3) and of $\mathcal{H}$; it is shown concretely by the theta decomposition in Section 4. Finally $D \equiv -\ell^2 \equiv \ell \pmod 2$. □

Theorem 3.4 shows in particular that the twined indices of magnetic charge one do not depend on the choice of strip, a fact also visible in the tables of [22]. For $g=1$ it recovers $d(n,\ell) = (-1)^{\ell+1}C_1^F(D)$ with $\Delta\psi_1^F = 3E_4A - 648\,\mathcal{H}$ [6].

[5] The polar part of B^2/A is $144\,A_{2,1}$ because $B(\tau,0)^2 = 144$, so the difference of the two sides is holomorphic in z. Adding to $\mathcal{H}$ Zagier's non-holomorphic correction makes it a Jacobi form of weight two and index one [24, 35], and the completion of $A_{2,1}$ has the opposite non-holomorphic part [6, Section 3]; hence the difference is a weak Jacobi form of weight two and index one for $\mathrm{SL}_2(\mathbb{Z})$, which by Lemma 4.1 below is a multiple of E_4A (since $M_2(\mathrm{SL}_2(\mathbb{Z})) = 0$), and comparing the coefficient of q^0y^1 shows that the multiple is zero. We have also checked the identity coefficient by coefficient for all exponents of q up to 99, expanding $A_{2,1}$ in the strip $-\operatorname{Im}\tau < \operatorname{Im} z < 0$.

4. The closed form

4.1. **Weak Jacobi forms of index one.** Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup. A weak Jacobi form of weight k and index one for Γ is a holomorphic function on $\mathbb{H}\times\mathbb{C}$ with the usual transformation laws under Γ and under $z \mapsto z+\lambda\tau+\mu$, whose Fourier expansion at every cusp contains only non-negative powers of q.

Lemma 4.1. *Let k be even and W a weak Jacobi form of weight k and index one for Γ, even in z. Then $W = f\,A + h\,B$ with $h = W(\tau,0)/12 \in M_k(\Gamma)$ and $f \in M_{k+2}(\Gamma)$. If $W = \sum_n \sum_\ell w(n,\ell) q^n y^\ell$ and $\mu_j(W) = \sum_n \big(\sum_\ell w(n,\ell)\ell^{2j}\big) q^n$, then*

$$h = \tfrac{1}{12}\,\mu_0(W), \qquad f = \tfrac{1}{2}\big(\mu_1(W) - 2E_2\,h\big). \tag{5}$$

Proof. The function $h(\tau) = W(\tau,0)/12$ is a modular form of weight k for Γ, holomorphic at the cusps since W is weak. The function $W - hB$ is a weak Jacobi form of weight k and index one, even in z, vanishing at $z=0$, hence vanishing to order two at every point of $\mathbb{Z}+\tau\mathbb{Z}$. Since A has exactly these zeros, each of order two, $f = (W-hB)/A$ is holomorphic, transforms like a Jacobi form of weight $k+2$ and index zero, and is therefore independent of z: a holomorphic function that is invariant under $z \mapsto z+\lambda\tau+\mu$ is bounded in z. So $f \in M_{k+2}(\Gamma)$; the behaviour at the cusps is controlled by (5), which expresses f through coefficients of W at each cusp. For (5), write $W = \sum_j \frac{(2\pi i z)^{2j}}{(2j)!}\mu_j(W)$, and use $A = (2\pi i z)^2 + O(z^4)$, $B = 12 + \mu_1(B)\frac{(2\pi i z)^2}{2} + O(z^4)$ with $\mu_1(B) = 2E_2$.[6] Comparing the coefficients of $(2\pi i z)^0$ and $(2\pi i z)^2/2$ gives $\mu_0(W) = 12h$ and $\mu_1(W) = 2f + 2E_2 h$. □

Proposition 4.2. *W_g is a weak Jacobi form of weight 2 and index one for $\Gamma(4N)$, even in z.*

Proof. Each of $\phi_g(\tau,z)^2$, $\phi_g(\tau/2,z)$, $\phi_g((\tau+1)/2,z)$ and $\phi_{g^2}(2\tau,2z)$ is a weak Jacobi form of weight 0 and index 2 for $\Gamma(4N)$. For the middle two, $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \Gamma(4N)$ acts on $\tau/2$ through $\left(\begin{smallmatrix} a & b/2 \\ 2c & d\end{smallmatrix}\right) \in \Gamma_0(N)$, and on $(\tau+1)/2$ through $\left(\begin{smallmatrix} a+c & (b+d-a-c)/2 \\ 2c & d-c\end{smallmatrix}\right) \in \Gamma_0(N)$; for the last, through $\left(\begin{smallmatrix} a & 2b \\ c/2 & d\end{smallmatrix}\right) \in \Gamma_0(N)$. At any cusp, $\gamma' \in \mathrm{SL}_2(\mathbb{Z})$ composed with $\tau \mapsto \tau/2$ factors as an element of $\mathrm{SL}_2(\mathbb{Z})$ followed by an upper triangular rational map, so the expansion at the cusp has only non-negative powers of q because ϕ_g is weak at all cusps [10, 19]. Hence $X_{g,2} = \frac{1}{2}\phi_g^2 + \frac{1}{2}\big(\phi_g(\frac{\tau}{2},z) + \phi_g(\frac{\tau+1}{2},z)\big) + \frac{1}{2}\phi_{g^2}(2\tau,2z)$ is weak of index two for $\Gamma(4N)$. Since $B^2 - E_4A^2$ is a weak Jacobi form of weight 0 and index 2 for $\mathrm{SL}_2(\mathbb{Z})$ and A one of weight -2 and index one, W_g transforms with weight 2 and index one and is holomorphic by Proposition 3.3. Its behaviour at the cusps follows from the moment formulas of Lemma 4.1 applied to $N_g = AW_g$: the Taylor coefficients of W_g in z are finite combinations of the moments of N_g and of the quasimodular series E_2, which have only non-negative powers of q at every cusp. □

Proof of Theorem B. By Lemma 4.1 and Proposition 4.2, $W_g = f\,A + h\,B$ with $f \in M_4(\Gamma(4N))$ and $h \in M_2(\Gamma(4N))$, and (5) computes the q-expansions of f and h from that of $X_{g,2}$, which in turn is given by Proposition 2.1 and (3). Both f and h have integral exponents of q, because U_2 and $\tau \mapsto 2\tau$ preserve integrality of exponents. Let f_g, h_g be the forms of Table 3; they lie in $M_4(\Gamma_0(N))$ and $M_2(\Gamma_0(N))$. By Sturm's theorem [34], a modular form of weight k for $\Gamma(M)$, $M \geq 3$, whose expansion in q vanishes to order greater than $\frac{k}{24}M^2\prod_{p\mid M}(1-p^{-2})$ is zero.[7] For $M = 4N \leq 32$ and $k = 4$ this order is at most 128. The q-expansions of $f - f_g$ and

[6] From $B = 12 + (2\pi i z)^2 E_2 + O(z^4)$; see [17, Section 9]. The coefficient of $(2\pi i z)^2/2$ in $fA + hB$ is $2f + 2E_2 h$.

[7] Sturm's bound is $\frac{k}{12}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(M)]/2$ in the local parameter $q^{1/M}$ at ∞, and $[\mathrm{PSL}_2(\mathbb{Z}) : \overline{\Gamma(M)}] = \frac{M^3}{2}\prod_{p\mid M}(1-p^{-2})$ for $M \geq 3$. Dividing by M converts to powers of q.

$h - h_g$ vanish beyond this order (the bound plus one for each type, which is q^{129} for $8A$) by exact rational arithmetic (Appendix C). Hence $f = f_g$ and $h = h_g$. Theorem 3.4 completes the proof. □

TABLE 3. The forms $f_g \in M_4(\Gamma_0(N))$ and $h_g \in M_2(\Gamma_0(N))$ of Theorem B, with $E_4^{(d)}(\tau) = E_4(d\tau)$.

g	f_g	h_g
$1A$	$3E_4$	0
$2A$	$\frac{7}{15}E_4 + \frac{64}{45}E_4^{(2)}$	$\frac{10}{9}\phi^{(2)}$
$3A$	$\frac{57}{160}E_4 + \frac{243}{160}E_4^{(3)}$	$\frac{9}{8}\phi^{(3)}$
$4B$	$\frac{7}{30}E_4 - \frac{14}{45}E_4^{(2)} + \frac{32}{15}E_4^{(4)}$	$-\frac{1}{18}\phi^{(2)} + \phi^{(4)}$
$5A$	$\frac{1}{32}E_4 - \frac{625}{288}E_4^{(5)} + \frac{25}{6}\big(\phi^{(5)}\big)^2$	$\frac{35}{36}\phi^{(5)}$
$6A$	$\frac{33}{160}E_4 - \frac{14}{45}E_4^{(2)} - \frac{117}{160}E_4^{(3)} + \frac{18}{5}E_4^{(6)} - \frac{1}{2}\phi^{(2)}\phi^{(3)}$	$-\frac{1}{18}\phi^{(2)} - \frac{1}{24}\phi^{(3)} + \frac{5}{6}\phi^{(6)}$
$7A$	$\frac{31}{960}E_4 - \frac{2401}{960}E_4^{(7)} + \frac{147}{32}\big(\phi^{(7)}\big)^2$	$\frac{7}{8}\phi^{(7)}$
$8A$	$\frac{7}{36}E_4^{(2)} - \frac{8}{15}E_4^{(4)} + \frac{64}{45}E_4^{(8)} + \frac{7}{6}\phi^{(2)}\phi^{(8)}$	$-\frac{1}{36}\phi^{(2)} + \frac{7}{9}\phi^{(8)}$

4.2. **Theta decomposition.** Let $\vartheta_{1,\mu}(\tau, z) = \sum_{r \equiv \mu\,(2)} q^{r^2/4} y^r$ and $\vartheta_\mu(\tau) = \vartheta_{1,\mu}(\tau, 0)$ for $\mu \in \{0, 1\}$, so that $\vartheta_0 = \sum_{s \in \mathbb{Z}} q^{s^2}$ and $\vartheta_1 = 2q^{1/4}\Theta(q)$ with $\Theta(q) = \sum_{k \geq 0} q^{k(k+1)}$. Every function $F = \sum c(n, \ell) q^n y^\ell$ with $c(n, \ell) = C(4n - \ell^2)$ can be written $F = F_0\vartheta_{1,0} + F_1\vartheta_{1,1}$ with $F_\mu(\tau) = \sum_{D \equiv -\mu^2\,(4)} C(D) q^{D/4}$. A direct computation, using $\eta(\tau)^6 = q^{1/4}(q)_\infty^6$ and the heat equation for theta functions, gives [6, Section 4]

$$A = \frac{-\vartheta_1\,\vartheta_{1,0} + \vartheta_0\,\vartheta_{1,1}}{\eta^6}, \qquad B = \frac{(24\vartheta_1' - E_2\vartheta_1)\,\vartheta_{1,0} + (E_2\vartheta_0 - 24\vartheta_0')\,\vartheta_{1,1}}{\eta^6},$$

$$\mathcal{H} = \mathcal{H}_0\,\vartheta_{1,0} + \mathcal{H}_1\,\vartheta_{1,1}, \qquad \mathcal{H}_0 = \sum_{n \geq 0} H(4n) q^n, \quad \mathcal{H}_1 = q^{-1/4}\mathcal{H}_1^\flat, \quad \mathcal{H}_1^\flat = \sum_{n \geq 1} H(4n-1) q^n,$$

where $' = q\,\frac{d}{dq}$. Writing $\vartheta_1' = 2q^{1/4}\Theta_1$ with $\Theta_1(q) = \sum_{k \geq 0}\big(k^2 + k + \frac{1}{4}\big) q^{k(k+1)}$, and using $\eta_g = qP_g$, we obtain from Theorem B the following explicit formulas. For $D = 4n'$, $n' \geq 1$,

$$d_g(4n') = [q^{n'+1}]\,\Big(\frac{2f_g\Theta - 2h_g\Theta_B}{(q)_\infty^6 P_g} + \frac{2d_g(2)\,\mathcal{H}_0}{P_g}\Big), \qquad \Theta_B = 24\,\Theta_1 - E_2\,\Theta, \tag{6}$$

and for $D = 4n' - 1$, $n' \geq 1$,

$$d_g(4n' - 1) = [q^{n'+1}]\,\Big(\frac{f_g\vartheta_0 + h_g\,(E_2\vartheta_0 - 24\vartheta_0')}{(q)_\infty^6 P_g} - \frac{2d_g(2)\,\mathcal{H}_1^\flat}{P_g}\Big). \tag{7}$$

Indeed, the $\vartheta_{1,0}$-component of $\eta_g\psi_g^F$ is $-2f_g\Theta/(q)_\infty^6 + 2h_g\Theta_B/(q)_\infty^6 - 2d_g(2)\mathcal{H}_0$, the coefficient of $q^{n'}$ of $\psi_{g,0}^F$ is $C_g^F(4n')$, and $d_g(4n') = -C_g^F(4n')$ by Theorem 3.4. The case $\mu = 1$ is the same with $d_g = +C_g^F$ and an extra factor $q^{-1/4}$ absorbed by ϑ_0 and $\mathcal{H}_1^\flat$. For $g = 1$, (6) and (7) become

$$d(4n') = [q^{n'+1}]\Big(\frac{6E_4\Theta}{(q)_\infty^{30}} + \frac{648\,\mathcal{H}_0}{(q)_\infty^{24}}\Big), \qquad d(4n' - 1) = [q^{n'+1}]\Big(\frac{3E_4\vartheta_0}{(q)_\infty^{30}} - \frac{648\,\mathcal{H}_1^\flat}{(q)_\infty^{24}}\Big),$$

the formulas of [6, Section 4]. Table 4 lists the first values. For $2A$, $3A$ and $4B$ they agree with the values of $-B_6^g$ in Tables 3, 6 and 9 of [22].

5. A majorant for the twined indices

5.1. **The comparison series.** For g of order $N \geq 2$ put, with all absolute values taken coefficientwise,

$$R_0 = \sum_{b=1}^{N-1} \big|2f_{g^b}\Theta - 2h_{g^b}\Theta_B\big|, \qquad R_1 = \sum_{b=1}^{N-1} \big|f_{g^b}\vartheta_0 + h_{g^b}(E_2\vartheta_0 - 24\vartheta_0')\big|, \tag{8}$$

and $H_\Sigma = \sum_{b=1}^{N-1} 2\, d_{g^b}(2)$. Define

$$k_0 = \frac{6E_4\Theta}{(q)_\infty^6} + 648\,\mathcal{H}_0 - R_0 - H_\Sigma\,|\mathcal{H}_0|, \tag{9}$$

$$k_1 = \frac{3E_4\vartheta_0}{(q)_\infty^6} - (648 + H_\Sigma)\,\mathcal{H}_1^\flat - R_1. \tag{10}$$

Proposition 5.1. *For g of order $N \geq 2$, $\mu \in \{0,1\}$ and $D = 4n' - \mu > 0$,*

$$d(D) - \sum_{b=1}^{N-1} \big|d_{g^b}(D)\big| \;\geq\; [q^{n'+1}]\,\frac{k_\mu(q)}{(q)_\infty^{24}}.$$

Proof. Take $\mu = 0$. For each b, write $F_b = 2f_{g^b}\Theta - 2h_{g^b}\Theta_B$. Since $1/(q^a;q^a)_\infty$ counts partitions into parts divisible by a, $0 \preceq 1/(q^a;q^a)_\infty \preceq 1/(q)_\infty$, and therefore $0 \preceq 1/P_{g^b} \preceq (q)_\infty^{-\sum_a m_a} \preceq (q)_\infty^{-16} \preceq (q)_\infty^{-18}$, using that $(q)_\infty^{-2}$ has non-negative coefficients and constant term one. Consequently

$$\Big|[q^k]\frac{F_b}{(q)_\infty^6 P_{g^b}}\Big| \leq [q^k]\frac{|F_b|}{(q)_\infty^6 P_{g^b}} \leq [q^k]\frac{|F_b|}{(q)_\infty^{24}}, \qquad \Big|[q^k]\frac{\mathcal{H}_0}{P_{g^b}}\Big| \leq [q^k]\frac{|\mathcal{H}_0|}{(q)_\infty^{24}}.$$

By (6), $|d_{g^b}(4n')| \leq [q^{n'+1}](|F_b| + 2d_{g^b}(2)|\mathcal{H}_0|)/(q)_\infty^{24}$, and summing over b and subtracting from the formula for $d(4n')$ gives the claim. The case $\mu = 1$ is identical, using (7) and $H(N) \geq 0$ for $N \geq 1$. □

5.2. **Coloured partitions.** The next lemma turns a coefficient inequality for k_μ into positivity of $k_\mu/(q)_\infty^{24}$.

Lemma 5.2. *For $k \geq 1$ and $n \geq 1$, $n\,p_k(n) = k\sum_{m=1}^{n} \sigma(m)\,p_k(n-m)$.*

Proof. Apply $q\frac{d}{dq}$ to $\log(q)_\infty^{-k} = k\sum_{m\geq 1}\sigma(m)q^m/m$, which follows from $-\log(1-q^j) = \sum_{r\geq 1} q^{jr}/r$. This gives $q\frac{d}{dq}(q)_\infty^{-k} = (q)_\infty^{-k}\cdot k\sum_m \sigma(m)q^m$; compare coefficients of q^n [1, Chapter 1]. □

Lemma 5.3. *Let $k(q) = \sum_{m\geq 0} a_m q^m$ with real coefficients and suppose that*

$$a_m > \frac{24\,|a_0|\,\sigma(m)}{\max(m,2)} \qquad \text{for all } m \geq 1. \tag{11}$$

Then $[q^n]\,k(q)/(q)_\infty^{24} > 0$ for every $n \geq 2$.

Proof. Let $n \geq 2$. For $1 \leq m \leq n$ we have $\max(m,2) \leq n$, so (11) gives $a_m > 24|a_0|\sigma(m)/n$. By Lemma 5.2 with $k = 24$,

$$\sum_{m=0}^{n} a_m p_{24}(n-m) > a_0\,p_{24}(n) + \frac{24|a_0|}{n}\sum_{m=1}^{n}\sigma(m)p_{24}(n-m) = a_0\,p_{24}(n) + |a_0|\,p_{24}(n) \geq 0.$$

The inequality is strict because $p_{24}(n-1) > 0$ and $a_1 > 24|a_0|/n$. □

Proposition 5.1 is applied with $n = n' + 1 \geq 2$, which is exactly the range $D \geq 3$ of Lemma 5.3.

6. Proof of Theorem C

By Proposition 5.1 and Lemma 5.3, Theorem C follows once (11) is established for the fourteen series k_μ, one for each type $g \neq 1$ and each $\mu \in \{0, 1\}$. Their constant terms a_0 are listed in Table 5; for $2A$ and $\mu = 1$ one has $a_0 = 0$, and (11) then says $a_m > 0$.

6.1. **Coefficient bounds.** Let c_A be the sum over $b = 1, \dots, N-1$ of the absolute values of the coefficients of f_{g^b} in the basis of Table 3, and c_B the same for h_{g^b} (Table 5).

Lemma 6.1. *For $i \geq 1$:*

(1) $|[q^i]\phi^{(d)}| \leq 24\,\sigma(i)$ *and* $|[q^i]E_2| \leq 24\,\sigma(i)$;
(2) *if F is $E_4(d\tau)$, a product $\phi^{(a)}\phi^{(b)}$, or a product $\phi^{(d)}E_2$, then* $|[q^i]F| \leq 240\,\sigma_3(i) \leq 289\,i^3$;
(3) $\sigma(i) \leq i\,(1 + \log i)$;
(4) $H(N) \leq 4i$ *for* $1 \leq N \leq 4i$.

All the series in (1) and (2) have constant term 1.

Proof. (1) $[q^i]\phi^{(d)} = 24\big(\sigma(i) - d\,\sigma(i/d)\big)/(d-1)$, with $\sigma(i/d) = 0$ if $d \nmid i$. Since $d\,\sigma(i/d) = \sum_{e|i/d} de$ is a sum over a subset of the divisors of i, $0 \leq \sigma(i) - d\sigma(i/d) \leq \sigma(i)$, and $d - 1 \geq 1$. (2) For $E_4(d\tau)$ the coefficient is $240\sigma_3(i/d) \leq 240\sigma_3(i)$. For a product FG of two series with constant term 1 and $|[q^j]F|, |[q^j]G| \leq 24\sigma(j)$,

$$|[q^i](FG)| \leq 48\sigma(i) + 576\sum_{j=1}^{i-1}\sigma(j)\sigma(i-j) = 48\sigma(i) + 48\big(5\sigma_3(i) + (1-6i)\sigma(i)\big) \leq 240\sigma_3(i),$$

by Ramanujan's identity $\sum_{j=1}^{i-1}\sigma(j)\sigma(i-j) = \frac{1}{12}\big(5\sigma_3(i) + (1-6i)\sigma(i)\big)$, which is the identity $E_2^2 = E_4 + 12E_2'$ read coefficientwise [35], and $96 \leq 288i$. Finally $\sigma_3(i) \leq \zeta(3)i^3 < 1.2021\,i^3$. (3) $\sigma(i)/i = \sum_{d|i} 1/d \leq \sum_{d=1}^{i} 1/d \leq 1 + \log i$. (4) Each reduced form $ax^2 + bxy + cy^2$ of discriminant $-N$ has $|b| \leq a \leq c$, hence $3a^2 \leq N$; for fixed a there are at most $a+1$ values of b, and c is then determined. So $H(N) \leq \sum_{a \leq \sqrt{N/3}}(a+1) \leq \frac{N}{6} + \frac{3}{2}\sqrt{N/3}$, which is at most $4i$ for $N \leq 4i$. □

Proposition 6.2. *Let $\mu_0 = 2$, $\mu_1 = 1$. For $m \geq 1$ the m-th coefficient a_m of k_μ satisfies*

$$a_m \;\geq\; \frac{\mu_\mu\, m^5}{40} - (2\sqrt{m}+2)\Big(289(c_A+c_B)m^3 + 576\,c_B\big(m+\tfrac{1}{4}\big)m(1+\log m)\Big) - 4m\big(648\,[\mu=1] + H_\Sigma\big).$$

Proof. Main term. Since $\Theta \succeq 1$ and $\vartheta_0 \succeq 1$ have non-negative coefficients, $[q^m]\,6E_4\Theta/(q)_\infty^6 \geq 6\,[q^m]E_4/(q)_\infty^6 \geq 6\big(p_6(m) + 240\sigma_3(m)\big)$, keeping only the two extreme terms of the convolution. Since $1/(1-q)^6 \preceq 1/(q)_\infty^6$, $p_6(m) \geq \binom{m+5}{5} > m^5/120$. So the main term of k_0 exceeds $m^5/20$ and that of k_1 exceeds $m^5/40$.

Class number terms. In k_0 the term $648\,\mathcal{H}_0$ has non-negative coefficients for $m \geq 1$ and is dropped; $H_\Sigma|\mathcal{H}_0|$ contributes at most $H_\Sigma H(4m) \leq 4mH_\Sigma$. In k_1 the coefficient of $(648+H_\Sigma)\mathcal{H}_1^b$ is $(648 + H_\Sigma)H(4m-1) \leq 4m(648 + H_\Sigma)$.

The series R_μ. Consider $\mu = 0$ and one b. The series 2Θ has non-negative coefficients summing to at most $2(\sqrt{m}+1)$ over exponents $\leq m$, since $k(k+1) \leq m$ forces $k \leq \sqrt{m}$. By Lemma 6.1(2), every coefficient of f_{g^b} of index $i \leq m$ is at most $289c_{A,b}m^3$ in absolute value (for

$i=0$ it is at most $c_{A,b}$), so $|[q^m](2f_{g^b}\Theta)| \le (2\sqrt{m}+2)\,289\,c_{A,b}\,m^3$. Likewise $|[q^m](2h_{g^b}E_2\Theta)| \le (2\sqrt{m}+2)\,289\,c_{B,b}\,m^3$. Finally $48\,h_{g^b}\Theta_1$ has coefficient

$$\Big|\sum_{k(k+1)\le m} 48\big(k^2+k+\tfrac14\big)[q^{m-k(k+1)}]h_{g^b}\Big| \le (\sqrt{m}+1)\cdot 48\big(m+\tfrac14\big)\cdot 24\,c_{B,b}\,m(1+\log m),$$

by Lemma 6.1(1),(3), since $|[q^i]h_{g^b}| \le 24c_{B,b}\sigma(i) \le 24c_{B,b}m(1+\log m)$ for $1\le i\le m$ and $|[q^0]h_{g^b}| \le c_{B,b}$. Summing over b gives the bound for $[q^m]R_0$. For $\mu=1$ the series ϑ_0 has coefficients summing to $2\lfloor\sqrt{m}\rfloor+1 \le 2\sqrt{m}+2$ over exponents $\le m$, and $24\vartheta_0'$ has coefficient $24s^2 \le 24(m+\frac14)$ at q^{s^2}; the same estimates apply. □

Corollary 6.3. *For every type and every μ, condition* (11) *holds for all $m \ge 11000$.*

Proof. By Lemma 6.1(3), $\sigma(m)/\max(m,2) \le 1+\log m$, so by Proposition 6.2 it suffices that

$$\frac{\mu_\mu}{40} > \frac{\rho(m)}{m^5},$$

where

$$\rho(m) = (2\sqrt{m}+2)\Big(289(c_A+c_B)m^3 + 576c_B\big(m+\tfrac14\big)m(1+\log m)\Big) + 4m\big(648[\mu=1]+H_\Sigma\big) + 24|a_0|(1+\log m).$$

The right side is a sum of terms $m^{-\alpha}(1+\log m)^\beta$, $m^{-\alpha}$ and $m^{-\alpha}(1+\frac{1}{4m})(1+\log m)$ with positive coefficients, $\alpha \ge 3/2$ and $\beta \le 1$; each is decreasing for $m\ge 1$, because $\frac{d}{dm}\log\big(m^{-\alpha}(1+\log m)\big) = \frac1m\big(-\alpha+\frac{1}{1+\log m}\big) < 0$. So the right side is decreasing, and it suffices to check the inequality at a single m_0. With the constants of Table 5 it holds at the values m_0 listed there, the largest being $m_0 = 10792$ for $7A$ and $\mu=1$; hence it holds for all $m \ge 11000$.[8] □

6.2. **The remaining coefficients.** For $1 \le m < 11000$ the coefficients a_m of the fourteen series k_μ are rational numbers that can be computed exactly from (8)–(10), Table 3 and the class numbers. We have computed all of them for $m \le 12045$ in exact rational arithmetic and checked (11) for each (Appendix C). The first coefficients are listed in Table 6. The smallest value of the ratio $a_m \max(m,2)/(24|a_0|\sigma(m))$ occurs at $m=1$ in every case and equals $2\cdot 1536/(24\cdot 96) = \frac43$ for $6A$ with $\mu=0$; the ratio grows faster than any power of m (Figure 4).

Proof of Theorem C. Let $D>0$ and write $D = 4n'-\mu$ with $\mu\in\{0,1\}$ and $n'\ge 1$ (the cases $D \equiv 1,2 \pmod 4$ do not occur). By Corollary 6.3 and the exact computation, k_μ satisfies (11). Lemma 5.3 with $n = n'+1 \ge 2$ gives $[q^{n'+1}]k_\mu/(q)_\infty^{24} > 0$, and Proposition 5.1 gives $d(D) > \sum_{b\ne 0}|d_{g^b}(D)|$. □

Proof of Theorem A. By Theorem 3.4, $d_{g^b}(n,\ell) = d_{g^b}(D)$ depends only on D. By (1) and Theorem C,

$$N\,S_a = d(D) + \sum_{b=1}^{N-1} e^{-2\pi i ab/N} d_{g^b}(D) \ge d(D) - \sum_{b=1}^{N-1}|d_{g^b}(D)| > 0.$$ □

[8] At $m=m_0$ the ratio of the two sides is between 1.000005 (for $5A$, $\mu=1$) and 1.0006; it was evaluated with forty significant digits. The thresholds are not optimised.

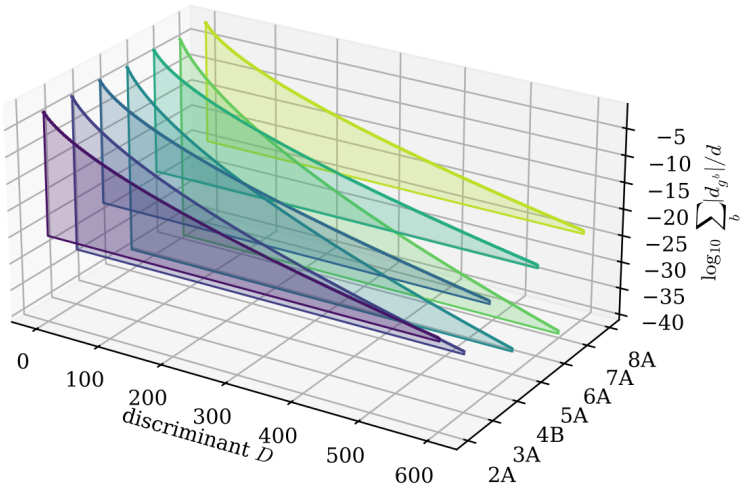


FIGURE 2. The ratio $\sum_{b\neq 0} |d_{g^b}(D)|/d(D)$ for $D \leq 600$, on a logarithmic scale. Theorem C says that it is below one for every $D > 0$. Its logarithm decreases roughly linearly in $\sqrt{D}$; the largest value is 0.052, for $6A$ at $D = 4$.

Remark 6.4. The argument proves more than positivity of S_a: every linear combination $\sum_b \lambda_b d_{g^b}$ with $|\lambda_b| \leq \lambda_0$ for $b \neq 0$ and $\lambda_0 > 0$ is positive. In particular the indices twined by characters of the cyclic group $\langle g \rangle$ and by sums of such characters are positive. The integrality of S_a is not used. We have checked that S_a is an integer for all $D \leq 600$ and every type; we do not prove integrality in general.

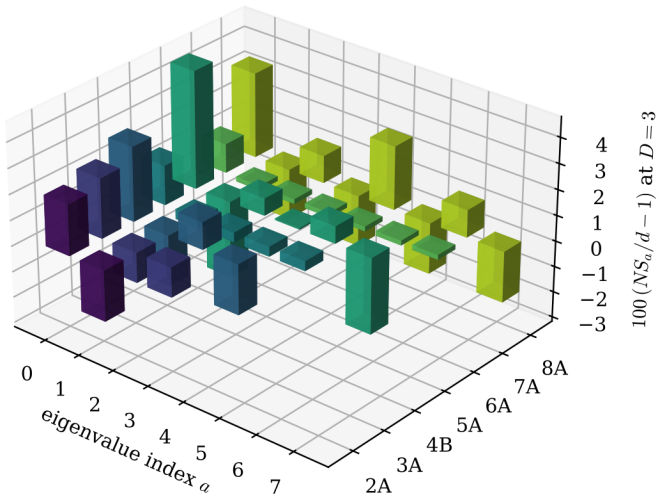


FIGURE 3. Deviation of the discrete information from equidistribution at the smallest discriminant $D = 3$: the bars show $100\,(NS_a/d - 1)$ for each type and each eigenvalue $e^{2\pi i a/N}$. All values lie between -3.1 and $+4.7$ per cent; at $D = 7$ their absolute values are below 1.1 per cent.

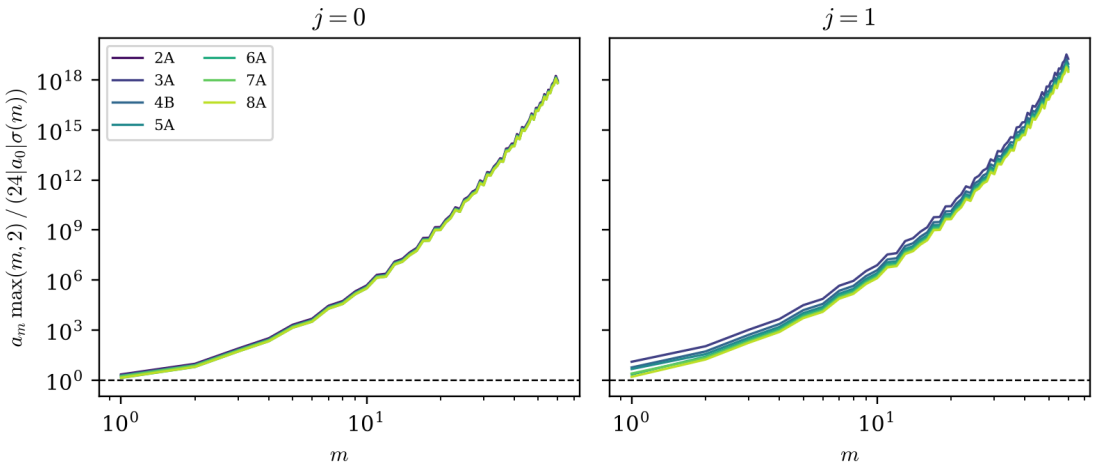


FIGURE 4. The ratio $a_m \max(m, 2)/(24|a_0|\sigma(m))$ for the series k_0 (left) and k_1 (right), $1 \le m \le 60$. Condition (11) requires the ratio to exceed one (dashed line). For $2A$ and $\mu = 1$ one has $a_0 = 0$ and the curve is omitted.

7. Discussion

To our knowledge, Theorem A is the first proof of the positivity of discrete information for an infinite family of charges. Three features of the proof are worth isolating. First, Lemma 3.1 and Theorem 3.4 show that for magnetic charge one Sen's attractor contour for twined indices

is equivalent to taking the Fourier coefficients of a single mock Jacobi form, uniformly in the twining. Second, Theorem B identifies this mock Jacobi form explicitly: the twining changes the holomorphic part $f_g A + h_g B$ and the eta quotient, while the mock part is always a multiple of the class number generating function $\mathcal{H}$, with coefficient $-2d_g(2)$ fixed by the residue at $z = 0$. Third, the twined indices are dominated by the untwined one at every charge, not only asymptotically.

For magnetic charge $P^2 = 4$ the polar part is again a multiple of an Appell–Lerch sum, and the decomposition of [6] for index two should allow the same argument; we have not carried it out. For larger magnetic charge the polar part has several components and the finite part is no longer determined by the residue at $z = 0$ alone; there the comparison of Theorem C would require bounds for the coefficients of the mock Jacobi forms $\psi^F_{g,m}$ of general index. Twisted-twining partition functions of the CHL orbifolds [22] and torsion larger than one are also not covered by our results.

Appendix A. The twining genus from the fixed points

Let g act on X with isolated fixed points $x_1, \dots, x_{\chi_g}$, and at x_j by $\mathrm{diag}(\zeta_j, \zeta_j^{-1})$, $\zeta_j = e^{2\pi i \alpha_j}$. By the holomorphic Lefschetz formula applied to the twisted Dolbeault complexes that define the elliptic genus, the g-equivariant elliptic genus localises to the fixed points [5, 23]:

$$\phi_g(\tau, z) = \sum_{j=1}^{\chi_g} \prod_{\epsilon=\pm 1} \frac{\vartheta_1(\tau, z + \epsilon\alpha_j)}{\vartheta_1(\tau, \epsilon\alpha_j)} = -\sum_{j=1}^{\chi_g} \frac{\vartheta_1(\tau, z+\alpha_j)\,\vartheta_1(\tau, z-\alpha_j)}{\vartheta_1(\tau, \alpha_j)^2},$$

where $\vartheta_1(\tau, z) = -iq^{1/8}(y^{1/2} - y^{-1/2}) \prod_{n\geq 1}(1-q^n)(1-q^n y)(1-q^n y^{-1})$. Each local term equals 1 at $z = 0$, so $\phi_g(\tau, 0) = \chi_g$. Using the product formula, with $c = \zeta + \zeta^{-1} = 2\cos 2\pi\alpha$,

$$-\frac{\vartheta_1(\tau, z+\alpha)\vartheta_1(\tau, z-\alpha)}{\vartheta_1(\tau, \alpha)^2} = \frac{y + y^{-1} - c}{2 - c} \prod_{n\geq 1} \frac{(1 - cq^n y + q^{2n} y^2)(1 - cq^n y^{-1} + q^{2n} y^{-2})}{(1 - cq^n + q^{2n})^2}. \tag{12}$$

Indeed $(y^{1/2}\zeta^{1/2} - y^{-1/2}\zeta^{-1/2})(y^{1/2}\zeta^{-1/2} - y^{-1/2}\zeta^{1/2}) = y + y^{-1} - c$, $(\zeta^{1/2} - \zeta^{-1/2})^2 = c - 2$, and $(1 - q^n y\zeta)(1 - q^n y\zeta^{-1}) = 1 - cq^n y + q^{2n} y^2$.

Proof of Proposition 2.1. Since ϕ_g is a weak Jacobi form of weight 0 and index one for $\Gamma_0(N)$, even in z, Lemma 4.1 gives $\phi_g = \frac{\phi_g(\tau,0)}{12} B + T\,A$ with $T \in M_2(\Gamma_0(N))$, and $\phi_g(\tau, 0) = \chi_g$. The space $M_2(\Gamma_0(N))$ has dimension at most 3 for $N \leq 8$, and Sturm's bound for weight two on $\Gamma_0(N)$ is $\frac{2}{12} N \prod_{p|N}(1 + \frac{1}{p}) \leq 2$. The coefficients of ϕ_g computed from (12) in the cyclotomic field $\mathbb{Q}(e^{2\pi i/N})$ agree exactly with those of $\frac{\chi_g}{12} B + T_g A$ for all exponents of q up to 20, far beyond this bound. Hence $T = T_g$. □

Appendix B. Exact values and constants

Table 4. The twined single-centred indices $d_g(D)$ for the smallest discriminants. For $2A$, $3A$ and $4B$ they agree with the entries $-B_6^g$ of Tables 3, 6 and 9 of [22] at $(Q^2, P^2) = (2,2), (2,4), (2,6)$ and $Q \cdot P \in \{0,1\}$, the tables being symmetric under $Q^2 \leftrightarrow P^2$.

g	$D=3$	$D=4$	$D=7$	$D=8$	$D=11$	$D=12$
$1A$	25353	50064	561576	1127472	8533821	16491600
$2A$	505	−1328	3128	−7696	14781	−33520
$3A$	297	−588	1272	−2442	4998	−8964
$4B$	117	−224	444	−768	1437	−2400
$5A$	98	−176	336	−588	1056	−1680
$6A$	37	−44	80	−130	246	−340
$7A$	48	−84	141	−228	387	−582
$8A$	27	−40	54	−96	165	−224

Table 5. Constants of Section 6: the sums c_A, c_B, H_Σ over the nontrivial powers of g, the constant terms a_0 of k_0 and k_1, and the points m_0 from which the analytic bound of Corollary 6.3 holds.

g	c_A	c_B	H_Σ	$a_0(k_0)$	$a_0(k_1)$	$m_0(k_0)$	$m_0(k_1)$
$2A$	1.889	1.111	104	−64	0	1089	1720
$3A$	3.750	2.250	108	−72	−3	1720	2719
$4B$	7.244	3.222	168	−80	−6	2483	3929
$5A$	25.472	3.889	112	−80	−9	4914	7784
$6A$	16.336	5.222	240	−96	−12	4005	6343
$7A$	42.763	5.250	108	−84	−15	6810	10792
$8A$	20.511	6.444	216	−96	−18	4645	7356

Table 6. The first coefficients $a_0, \ldots, a_3$ of the comparison series k_0 and k_1 of Section 5.

	k_0				k_1			
g	a_0	a_1	a_2	a_3	a_0	a_1	a_2	a_3
$2A$	−64	1652	21434	$\frac{470896}{3}$	0	$\frac{1286}{3}$	11205	99176
$3A$	−72	1710	21891	159672	−3	444	11403	98952
$4B$	−80	1620	21210	156880	−6	406	11109	98792
$5A$	−80	1704	22140	$\frac{478856}{3}$	−9	488	11497	99416
$6A$	−96	1536	20844	156640	−12	$\frac{926}{3}$	10813	98024
$7A$	−84	1674	22191	159564	−15	438	11583	99546
$8A$	−96	1564	21110	156720	−18	334	10965	98552

Appendix C. Computations

Three finite computations enter the proofs, all in exact rational arithmetic with the FLINT library: (i) the coefficients of ϕ_g in Proposition 2.1 up to q^{20}, in exact arithmetic in $\mathbb{Q}(e^{2\pi i/N})$; (ii) the expansions of f and h in Theorem B from (3) and (5), compared with Table 3 up to the Sturm bound q^{129} for $\Gamma(4N)$; (iii) the coefficients a_m of the fourteen series k_μ for $m \le 12045$ and condition (11). The class numbers are computed by counting reduced forms. The identity of Proposition 3.2 was checked in the same way up to q^{99}. As an independent check, S_a was computed for all $D \le 600$ and every type from the theta components of Theorem B; every S_a is a positive integer, and the values for $N = 2, 3, 4$ agree with [22]. The programs and their output are archived at [4].

Declarations

Data availability. The programs that carry out the computations of Appendix C and their output are available in the archive [4].
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Conflicts of interest. The author declares no conflict of interest.
Ethics. Not applicable; this work involves no human or animal subjects.

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